

Redis Advanced Interview Preparation Guide

1. Redis Cluster

What is Redis Cluster?

Redis Cluster is used for horizontal scaling and high availability. Data is distributed across multiple Redis nodes.

Horizontal Scaling:

Instead of storing all data in one server, Redis distributes data across multiple servers.

Sharding Basics:

Sharding means splitting data into smaller parts and storing them in different nodes.

Example:

User 1-1000 -> Node A

User 1001-2000 -> Node B

Interview Answer:

"Redis Cluster use pannu horizontal scaling and fault tolerance achieve pannuvom. Data sharding use pannu multiple nodes la distribute aagum."

2. Replication & Sentinel

Master-Slave Replication:

One master node handles writes and slave nodes replicate data from master.

High Availability:

If one server fails, another replica can continue serving requests.

Failover Basics:

Redis Sentinel monitors Redis instances and automatically promotes a slave as master if the master fails.

Interview Answer:

"Redis replication and Sentinel use pannu high availability and automatic failover implement pannuvom."

3. Memory Optimization

Why Redis Fast?

Redis stores data in memory (RAM), so data access is extremely fast.

In-Memory Architecture:

Data is stored directly in RAM instead of disk-based storage.

Single-Threaded Event Loop:

Redis primarily uses a single-threaded event loop to process commands efficiently without thread locking overhead.

Interview Answer:

"Redis in-memory architecture and single-threaded event loop nala extremely fast ah work pannum."

4. Pub/Sub vs Streams

Pub/Sub:

Real-time messaging system without message persistence.
If subscriber offline na messages miss aagum.

Streams:

Persistent log-based messaging system with consumer groups and acknowledgements.

Main Difference:

Pub/Sub -> Real-time only

Streams -> Persistent event processing

Interview Answer:

"Pub/Sub real-time communication ku use pannuvom, Streams reliable event processing ku use pannuvom."

5. BullMQ Advanced Concepts

Retry Jobs:

Failed jobs automatically retry panna configure pannalam.

Delayed Jobs:

Specific delay aprm jobs execute panna mudiyum.

Failed Jobs:

Failed jobs separate ah track pannalam and retry pannalam.

Interview Answer:

"BullMQ use panni retry mechanisms, delayed processing, and failed job handling implement pannuvom."

6. Cache Patterns

Cache Aside:

Application first cache check pannum. Cache miss na DB fetch panni cache update pannum.

Write Through:

Data DB and cache rendu place um same time update aagum.

Write Back:

Data first cache la write aagum, later DB ku sync aagum.

Interview Answer:

"Different cache strategies use panni performance and consistency maintain pannuvom."

7. Real-Time Use Cases

Chat Applications:

Pub/Sub use panni real-time chat messages send pannuvom.

Notifications:

Redis queues and Pub/Sub use panni notifications process pannuvom.

OTP Systems:

TTL use panni OTP automatic expiration implement pannuvom.

Payment Lock:

Distributed locking use panni duplicate payments avoid pannuvom.

Queue Processing:

BullMQ use panni background jobs process pannuvom.

8. Final Interview Tips

- Always explain with real-world examples.
- Mention Redis is an in-memory datastore.
- Mention Redis is extremely fast.
- Explain practical use cases.
- Mention TTL, caching, BullMQ, and distributed locking.
- Explain Pub/Sub and Streams difference clearly.